# One goo, two ways

Cornstarch Goo

Name _____ Date _____



Little Wonderer stirs white powder into a bowl of water. The goo looks runny, like milk. A slow mitten sinks right in. Then the little one taps it fast. Tap! Now it feels as hard as a table. Big Wonderer leans in to look. How can one bowl be soft and hard?



Try two spoons. Rest one spoon on the goo and wait. It sinks slowly to the bottom. Now smack the goo with the other spoon. It stops at the top and leaves a dull dent! Most liquids act the same, fast or slow. Water does, and so does honey. This goo breaks that rule, so it is called non-Newtonian.



Cornstarch is made of tiny grains. About five could sit across one hair. In the goo, thin water sits between them. Push slowly, and the water slips aside. The grains slide past like skaters. Push fast, and the water cannot move in time. The grains crash together and lock into chains. That is called jamming.

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|---|---|
| **non-Newtonian** | a liquid whose thickness changes when you push it harder or faster |
| **grain** | a very small, hard bit of something, like cornstarch or sand |
| **jamming** | when crowded grains lock together and stop sliding |
| **liquid** | something that flows and takes the shape of its bowl |
| **solid** | something that keeps its own shape |

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Here is a clue. Squeeze a ball of goo, and its skin turns dull and cracked. It looks almost dry! Jammed grains need a little more room, so they poke up through the water. Stop squeezing, and the grains settle back. The water covers them again, and the ball melts into a shiny puddle.



Back at the table, Big Wonderer lifts a spoonful. A slow ribbon drips back into the bowl. Little Wonderer reaches up to it. Now the guess changes. The goo is not two things. It is one crowd of grains in water. Gentle pushes let them slide. Quick pushes jam them, and then the jam melts away.



Wet sand at the beach plays a cousin of this trick. Step on firm, wet sand, and it turns pale around your foot. Your step pushes the grains apart, so water sinks into the new gaps. Osborne Reynolds explained it in 1885. People have even run across a pool of goo. If they stop, they sink. How long can a jam last?

1. What happens when you tap the goo fast?
2. Why does a resting spoon sink but a smacked spoon stop?
3. Why does a squeezed ball of goo look dry?
4. Why does wet sand turn pale around a foot?

## Your turn

Write or dictate three sentences: what a slow push does, what a fast push does, and what the tiny grains are doing.

Example: A slow push sinks in. A fast push feels hard. The grains jam when the water cannot move out of the way.